ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. FOV $= W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

Per pixel $= 160/1280 \approx$ **0.13 mm/px**.

(b) LSB $= 5/4096 = 1.2207$ mV.

$\Delta F =$ LSB / S $= 1.2207$ mV / 10 mV/N = **0.0122 N**.

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$ for a ZOH input.

$F = e^{-0.2} =$ **0.82**; $G = (0.82 - 1)/(-2) =$ **0.09**. So $x_{k+1} = 0.82\, x_k + 0.09\, u_k$.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\, x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Since $|\sin\theta| \le 1$, an equilibrum exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6 \times 0.5,\ -2.5] =$ **[0, 1; −61, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$.

At θ_d the left side is 0 but the right side is 2.4525, a contradiction, so $\theta_{ss} \neq \theta_d$ (graviy needs a holding torque).

Solving $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$ numerically: $\theta_{ss} \approx 87.2°$, error ≈ **2.8°**.

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensation the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90°$ $\cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\, r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$.

With $a = 2$, $b = 1$, $c = 6$, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} =$ **0.8**. Tracking error $= (1 - 0.8)/1 \times 100 =$ **20%**. (Less than the 25% acuator loss, because the stiffness also dropped.)

(c) $\ddot{\theta} + a\dot{\theta} + c\theta = b(-k_1\theta - k_2\dot{\theta} + K_I\xi)$, $\dot{\xi} = r - \theta$. Take $r = 0$ and differentiate once, using $\dot{\xi} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\ddot{\theta} + (c + bk_1)\dot{\theta} + bK_I\theta = 0$.

Char. poly: **$s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$**.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; $K_I = 32$ ($b = 1$).

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the acuator saturtes the error persists, ξ keeps growing and causes a large overshoot afterwards. Remidy: anti-windup (clamp the integrator, conditional integration or back-calculation).